SAMANTHA (SAM) LUI

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EDUCATION

Bachelor of Computer Science (Third Year) | University of British Columbia *September 2024 – December 2027*

Advanced Diploma – Clinical Genetics (88%) | British Columbia Institute of Technology *August 2020 – October 2021*

Bachelor of Health Sciences (Dean's Honour Roll) | Simon Fraser University *September 2011 – December 2017*

PROFESSIONAL EXPERIENCE

UBC Faculty of Applied Sciences – Front-End Web Developer (Incoming) | Vancouver, B.C. *Starting May 2026*

- Selected to support the development and maintenance of web marketing content using **HTML**, **CSS**, **JavaScript** and **CMS** platforms, with a focus on responsive design, accessibility, and usability best practices

CIHI – Applied Cloud Architecture Developer Intern | North York, O.N. *January 2026 – April 2026*

- Migrated 3 enterprise **Java** monoliths from Spring to **Spring Boot**, refactoring 20+ **Maven** dependencies and converting XML configurations (I.e., **Spring Security**) to Java-based configs, improving cloud deployment stability on **Amazon EKS**
- Resolved containerization failures by replacing legacy **JSP** views with **Thymeleaf**, and automated **Docker** image builds with **Bamboo** CI/CD, reducing build and debugging time by **> 90%**
- Improved overall test coverage across applications from **76%** to **88%** using **Mockito** and **JUnit Jupiter** with the addition of more comprehensive unit tests, improving confidence in application behaviour during migration
- Built a responsive internal knowledge repository using **HTML5**, **CSS3**, and **JavaScript**, outlining migration and troubleshooting workflows to improve team onboarding efficiency and knowledge transfer for a **20+** member team
- Contributed to an **AWS Bedrock proof-of-concept** to evaluate the usage of a multi-agent and human-in-the-loop validation workflow for automated logical data modelling

Shared Services Canada – I.T. Analyst Co-op Student | Vancouver, B.C. *August 2025 – December 2025*

- Delivered **Windows 11** imaging, and managed **Microsoft Azure** user states with **MDM** and **SCCM**, leveraging **PowerShell** automation for secure account provisioning and system performance optimization
- Developed an internal automation tool with **Python**, **Tkinter** and **Selenium**, streamlining workflows with reduced manual data entry errors and Excel template auto-population, improving ticket completion speed by **75%**
- Authored a **60+** page training guide standardizing support procedures, adopted as a core onboarding resource for new co-ops across regional support teams

TECHNICAL PROJECTS

PassPing | *React.js, Vite, TypeScript, Chrome APIs* | GitHub *March 2026 – Present*

- Building a Chrome extension using **Manifest V3** with event-driven background service workers and **chrome.alarms** to automate time-based reminders, persisting user state with **chrome.storage** to reduce missed student pass renewals with user-configurable tracking logic

OverHacked! | *Phaser.js, Next.js, FastAPI, LangChain* | GitHub *January 2026 – January 2026*

- Built **FastAPI** backend endpoints to support agentic AI workflows, integrating OpenAI LLMs with **CrewAI** to dynamically generate topic-specific programming quizzes for mini-game interactions

AniMori – Anime Library App | *Electron.js, React, Vite, SQLite, Node.js, Jikan* | GitHub *August 2025 – December 2025*

- Developed a cross-platform **Electron.js**, **React** + **SQLite**-backed desktop app with automated metadata retrieval via the **Jikan API,** with **IPC** channel implementation for client-to-backend connectivity

JobQuest AI – AI-Powered Job App Assistant | *Next.js, AWS Bedrock, ElevenLabs* | GitHub *October 2025 – October 2025*

- Designed a gamified job search assistant with a **Next.js** video-game style UI, increasing engagement via interactive resume reviews and mock behavioural interviews with AI-powered features (**Bedrock**, **ElevenLabs**) for personalized insights

TECHNICAL SKILLS

Languages | Python, Java, C++, PowerShell (currently acquiring), SQL, R, HTML5, CSS3, JavaScript

Frameworks/Libraries | React.js, Node.js/Express.js, Next.js, Electron.js, Spring Boot, JUnit Jupiter, Mockito, FastAPI

Databases | PostgreSQL, MongoDB (NoSQL), SQLite, H2

Cloud & Containerization Platforms | Docker, JFrog Artifactory, Kubernetes, Amazon Web Services (ECR, EKS)

Developer Tools | Git/GitHub, GitLab, Atlassian (Confluence, Bamboo, Jira), Netlify, Postman, Vite, Jupyter, Groq, Selenium

Enterprise Systems | Microsoft Azure (AD, Mobile Device Management, Microsoft Endpoint Configuration Manager)